

# Markdown PDF

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This extension converts Markdown files to pdf, html, png or jpeg files.

[Japanese README](#)

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## Breaking Changes in 2.0.0

Version 2.0.0 introduces changes that may affect existing behavior. See the [FAQ](#) section for details.

- Heading IDs now follow GitHub-compatible VS Code slug generation. Existing internal anchors in your documents may change. See [Why did my heading anchors change?](#)
- Highlight.js upgraded from v9 to v11. Some highlight style names have been renamed or removed. See [Why did my syntax highlight style stop working?](#)
- Front matter parsing is now stricter. Some previously accepted formats may be rejected. See [Why is my front matter no longer parsed?](#)
- Chromium is resolved from an installed Chrome/Edge browser first, or auto-downloaded on first use. See [How is the Chromium browser selected?](#) and [Where is Chromium downloaded?](#)

## Features

| Feature                             | Description                                                | Example                           |
|-------------------------------------|------------------------------------------------------------|-----------------------------------|
| <a href="#">Syntax highlighting</a> | Code block highlighting via highlight.js                   | <code>```js</code>                |
| <a href="#">Emoji</a>               | Emoji shortcodes                                           | <code>:smile:</code>              |
| Checkbox                            | GitHub-style task lists (built-in custom plugin)           | <code>- [ ] / - [x]</code>        |
| Heading IDs                         | GitHub-compatible heading anchors (built-in custom plugin) | <code># Heading → #heading</code> |
| <a href="#">Container</a>           | Admonition-like blocks                                     | <code>::: warning</code>          |

| Feature  | Description                                       | Example               |
|----------|---------------------------------------------------|-----------------------|
| Include  | Embed Markdown fragments (built-in custom plugin) | : [Label] (path.md)   |
| PlantUML | UML diagrams from code blocks                     | @startuml ... @enduml |
| Mermaid  | Diagrams from fenced code blocks                  | ```mermaid            |

Sample files

- pdf
- html
- png
- jpeg

Heading IDs

Headings automatically receive GitHub-compatible anchor IDs. For example:

| Heading         | Generated ID   |
|-----------------|----------------|
| # My Heading    | #my-heading    |
| # API Reference | #api-reference |
| # 日本語見出し        | #日本語見出し        |

See [Why did my heading anchors change?](#) in the FAQ for details.

Checkbox

INPUT

```
- [ ] Task A
- [x] Task B
```

OUTPUT

```
<ul>
  <li><input type="checkbox" disabled> Task A</li>
  <li><input type="checkbox" disabled checked> Task B</li>
</ul>
```

Container

INPUT

```
::: warning
*here be dragons*
:::
```

## OUTPUT

```
<div class="warning">
<p><em>here be dragons</em></p>
</div>
```

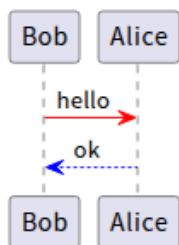
## PlantUML

Render UML diagrams via [PlantUML](#) using [markdown-it-plantuml](#).

## INPUT

```
@startuml
Bob -[#red]> Alice : hello
Alice -[#0000FF]->Bob : ok
@enduml
```

## OUTPUT



## Include

Include markdown fragment files: `:\[alternate-text\]\(relative-path-to-file.md\)`.

```
|—— [plugins]
|   |—— README.md
|—— CHANGELOG.md
|—— README.md
```

## INPUT

README Content

`:\[Plugins\]\(../plugins/README.md\)`

`:\[Change log\]\(CHANGELOG.md\)`

## OUTPUT

Content of README.md

Content of plugins/README.md

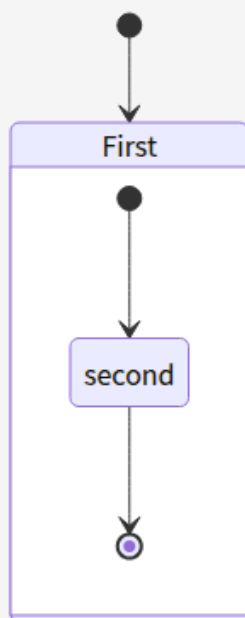
Content of CHANGELOG.md

## Mermaid

### INPUT

```
```mermaid
stateDiagram
    [*] --> First
    state First {
        [*] --> second
        second --> [*]
    }
...
```
```

### OUTPUT



# Chromium

Markdown PDF uses a Chromium-based browser for PDF/PNG/JPEG export. It tries the following sources in order:

1. The path specified in [markdown-pdf.executablePath](#)
2. An installed Google Chrome, Microsoft Edge, or Chromium on your system
3. A managed Chromium automatically downloaded on first use

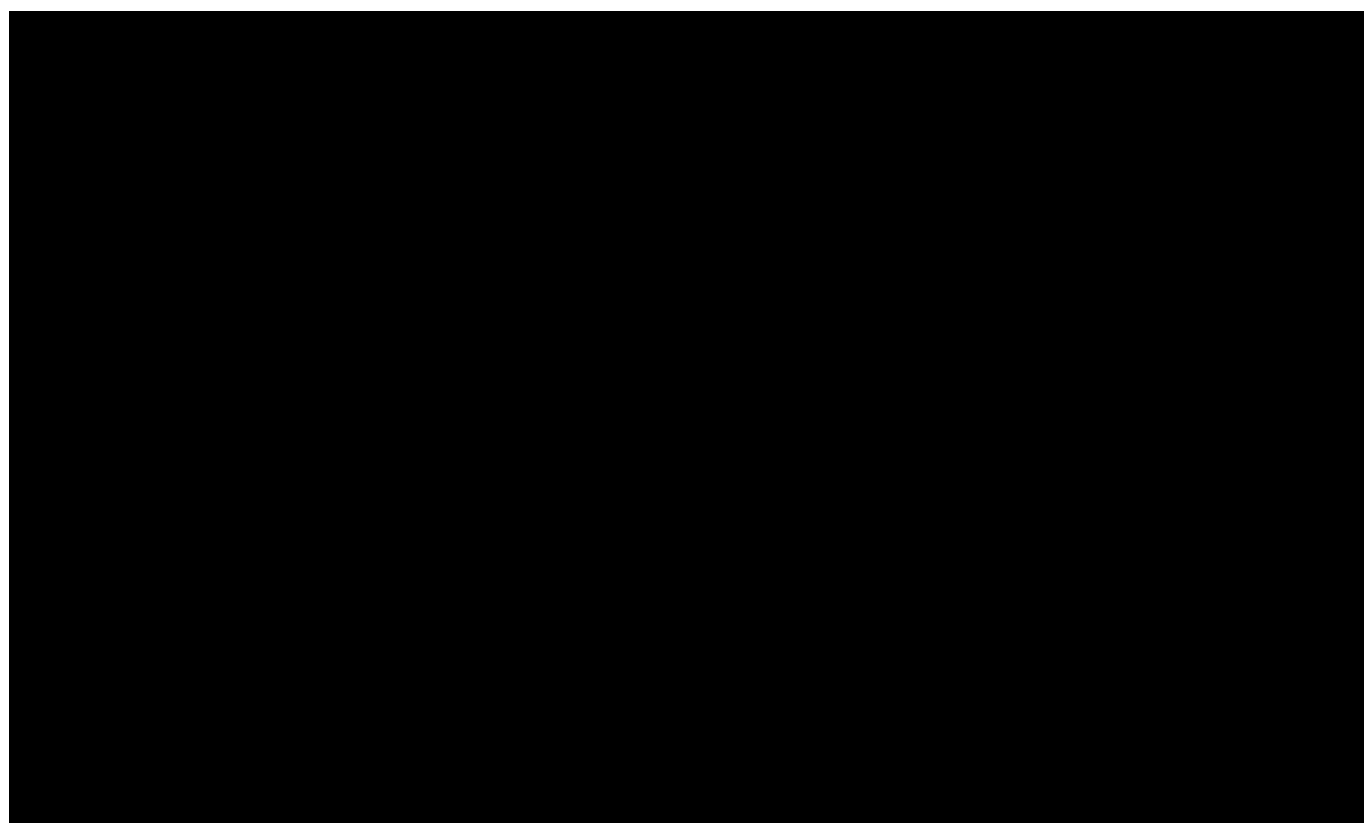
See [How is the Chromium browser selected?](#) and [Where is Chromium downloaded?](#) in the FAQ for details.

If you are behind a proxy, set the [http.proxy](#) option in settings.json and restart Visual Studio Code.

## Usage

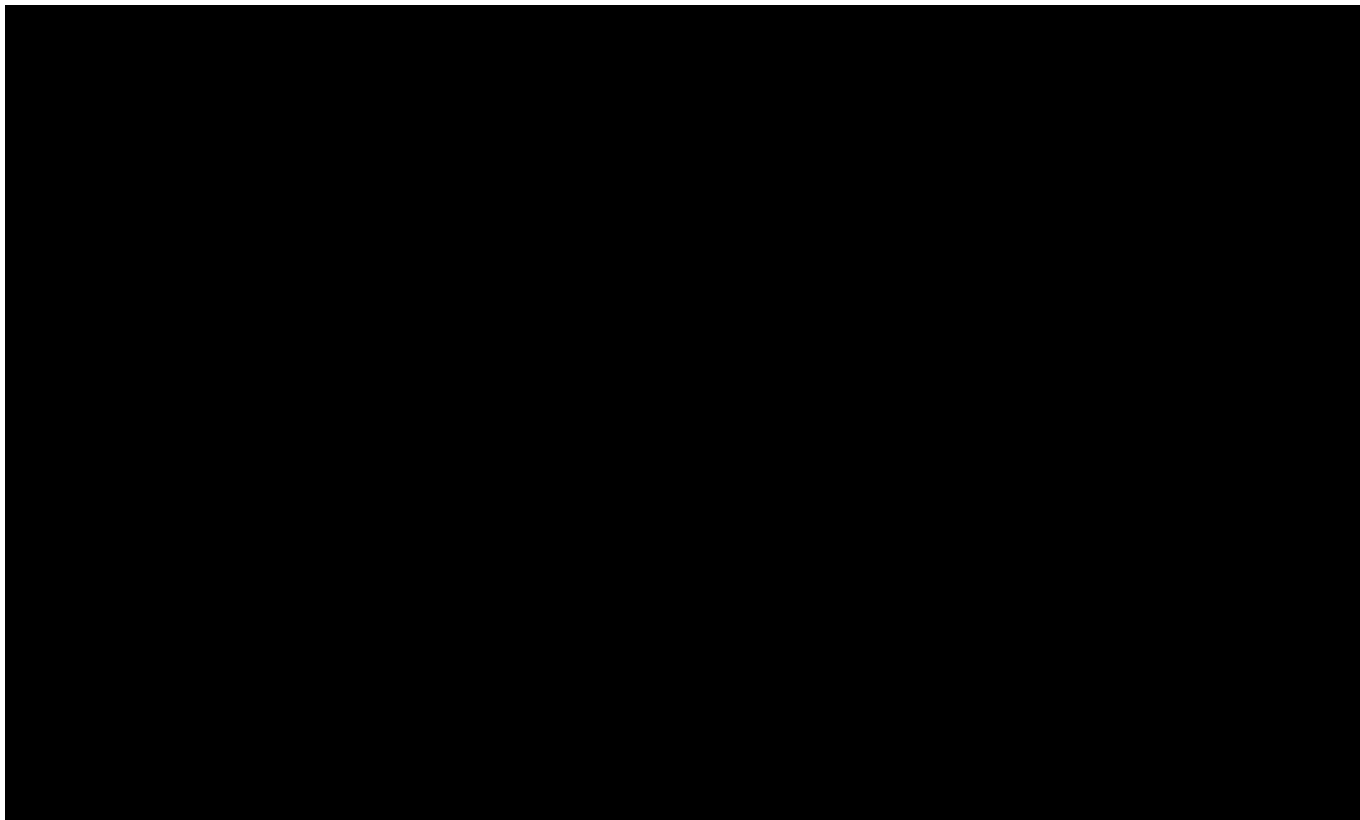
### Command Palette

1. Open the Markdown file
2. Press **F1** or **Ctrl+Shift+P**
3. Type **export** and select below
  - markdown-pdf: Export (settings.json)
  - markdown-pdf: Export (pdf)
  - markdown-pdf: Export (html)
  - markdown-pdf: Export (png)
  - markdown-pdf: Export (jpeg)
  - markdown-pdf: Export (all: pdf, html, png, jpeg)



### Menu

1. Open the Markdown file
2. Right click and select below
  - markdown-pdf: Export (settings.json)
  - markdown-pdf: Export (pdf)
  - markdown-pdf: Export (html)
  - markdown-pdf: Export (png)
  - markdown-pdf: Export (jpeg)
  - markdown-pdf: Export (all: pdf, html, png, jpeg)



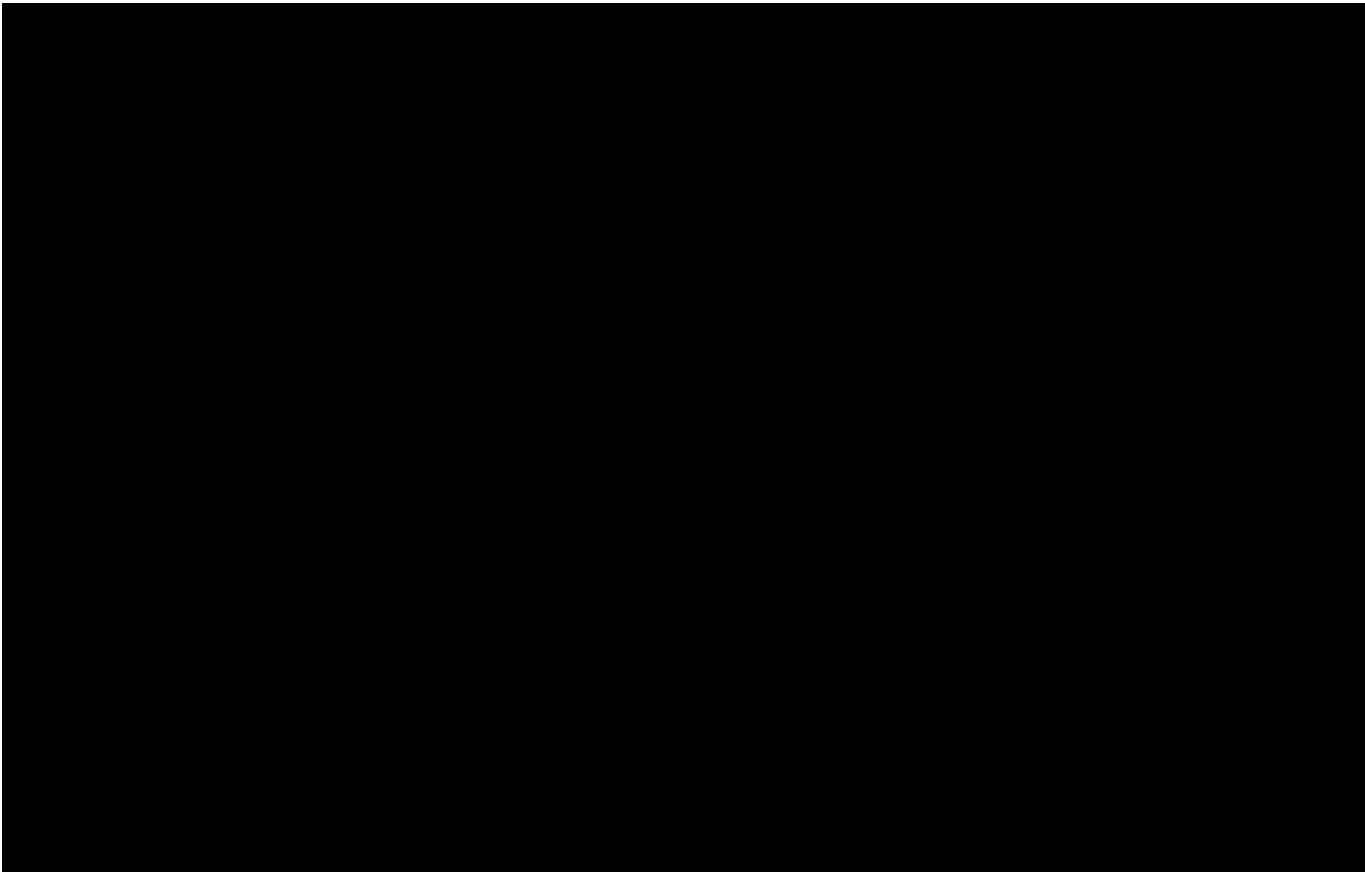
## Auto convert

1. Add "markdown-pdf.convertOnSave": true option to **settings.json**
2. Restart Visual Studio Code
3. Open the Markdown file
4. Auto convert on save

## Extension Settings

### Visual Studio Code User and Workspace Settings

1. Select **File > Preferences > UserSettings or Workspace Settings**
2. Find markdown-pdf settings in the **Default Settings**
3. Copy **markdown-pdf.\*** settings
4. Paste to the **settings.json**, and change the value



# Options

List

| Category                 | Option name                                  | Configuration scope |
|--------------------------|----------------------------------------------|---------------------|
| Save options             | markdown-pdf.type                            |                     |
|                          | markdown-pdf.convertOnSave                   |                     |
|                          | markdown-pdf.convertOnSaveExclude            |                     |
|                          | markdown-pdf.outputDirectory                 |                     |
|                          | markdown-pdf.outputDirectoryRelativePathFile |                     |
| Styles options           | markdown-pdf.styles                          |                     |
|                          | markdown-pdf.stylesRelativePathFile          |                     |
|                          | markdown-pdf.includeDefaultStyles            |                     |
| Syntax highlight options | markdown-pdf.highlight                       |                     |
|                          | markdown-pdf.highlightStyle                  |                     |
| Markdown options         | markdown-pdf.breaks                          |                     |
| Emoji options            | markdown-pdf.emoji                           |                     |
| Configuration options    | markdown-pdf.executablePath                  |                     |



| Category                    | Option name                             | Configuration scope |
|-----------------------------|-----------------------------------------|---------------------|
| Common Options              | markdown-pdf.scale                      |                     |
| PDF options                 | markdown-pdf.displayHeaderFooter        | resource            |
|                             | markdown-pdf.headerTemplate             | resource            |
|                             | markdown-pdf.footerTemplate             | resource            |
|                             | markdown-pdf.printBackground            | resource            |
|                             | markdown-pdf.orientation                | resource            |
|                             | markdown-pdf.pageRanges                 | resource            |
|                             | markdown-pdf.format                     | resource            |
|                             | markdown-pdf.width                      | resource            |
|                             | markdown-pdf.height                     | resource            |
|                             | markdown-pdf.margin.top                 | resource            |
|                             | markdown-pdf.margin.bottom              | resource            |
|                             | markdown-pdf.margin.right               | resource            |
|                             | markdown-pdf.margin.left                | resource            |
| PNG JPEG options            | markdown-pdf.quality                    |                     |
|                             | markdown-pdf.clip.x                     |                     |
|                             | markdown-pdf.clip.y                     |                     |
|                             | markdown-pdf.clip.width                 |                     |
|                             | markdown-pdf.clip.height                |                     |
|                             | markdown-pdf.omitBackground             |                     |
| PlantUML options            | markdown-pdf.plantumlOpenMarker         |                     |
|                             | markdown-pdf.plantumlCloseMarker        |                     |
|                             | markdown-pdf.plantumlServer             |                     |
| markdown-it-include options | markdown-pdf.markdown-it-include.enable |                     |
| mermaid options             | markdown-pdf.mermaidServer              |                     |

## Save options

### markdown-pdf.type

- Output format: pdf, html, png, jpeg
- Multiple output formats support
- Default: pdf

```
"markdown-pdf.type": [  
  "pdf",  
  "html",  
  "png",  
  "jpeg"  
],
```

### markdown-pdf.convertOnSave

- Enable Auto convert on save
- boolean. Default: false
- To apply the settings, you need to restart Visual Studio Code

### markdown-pdf.convertOnSaveExclude

- Excluded file name of convertOnSave option

```
"markdown-pdf.convertOnSaveExclude": [  
  "^work",  
  "work.md$",  
  "work|test",  
  "[0-9][0-9][0-9][0-9]-work",  
  "work\\test" // All '\\' need to be written as '\\' (Windows)  
],
```

### markdown-pdf.outputDirectory

- Output Directory
- All \ need to be written as \\ (Windows)

```
"markdown-pdf.outputDirectory": "C:\\work\\output",
```

- Relative path
  - If you open the **Markdown file**, it will be interpreted as a relative path from the file
  - If you open a **folder**, it will be interpreted as a relative path from the root folder
  - If you open the **workspace**, it will be interpreted as a relative path from each root folder
    - See [Multi-root Workspaces](#)

```
"markdown-pdf.outputDirectory": "output",
```

- Relative path (home directory)
  - If path starts with ~, it will be interpreted as a relative path from the home directory

```
"markdown-pdf.outputDirectory": "~/output",
```

- If you set a directory with a **relative path**, it will be created if the directory does not exist
- If you set a directory with an **absolute path**, an error occurs if the directory does not exist

### markdown-pdf.outputDirectoryRelativePathFile

- If `markdown-pdf.outputDirectoryRelativePathFile` option is set to **true**, the relative path set with `markdown-pdf.outputDirectory` is interpreted as relative from the file
- It can be used to avoid relative paths from folders and workspaces
- boolean. Default: false

## Styles options

### markdown-pdf.styles

- A list of local paths to the stylesheets to use from the markdown-pdf
- If the file does not exist, it will be skipped
- All `\` need to be written as `\\` (Windows)

```
"markdown-pdf.styles": [
  "C:\\Users\\<USERNAME>\\Documents\\markdown-pdf.css",
  "/home/<USERNAME>/settings/markdown-pdf.css",
],
```

- Relative path
  - If you open the **Markdown file**, it will be interpreted as a relative path from the file
  - If you open a **folder**, it will be interpreted as a relative path from the root folder
  - If you open the **workspace**, it will be interpreted as a relative path from each root folder
    - See [Multi-root Workspaces](#)

```
"markdown-pdf.styles": [
  "markdown-pdf.css",
],
```

- Relative path (home directory)
  - If path starts with `~`, it will be interpreted as a relative path from the home directory

```
"markdown-pdf.styles": [
  "~/.config/Code/User/markdown-pdf.css"
],
```

- Online CSS (<https://xxx/xxx.css>) is applied correctly for JPG and PNG, but problems occur with PDF [#67](#)

```
"markdown-pdf.styles": [  
  "https://xxx/markdown-pdf.css"  
],
```

### markdown-pdf.stylesRelativePathFile

- If `markdown-pdf.stylesRelativePathFile` option is set to `true`, the relative path set with `markdown-pdf.styles` is interpreted as relative from the file
- It can be used to avoid relative paths from folders and workspaces
- boolean. Default: false

### markdown-pdf.includeDefaultStyles

- Enable the inclusion of default Markdown styles (VSCode, markdown-pdf)
- boolean. Default: true

## Syntax highlight options

### markdown-pdf.highlight

- Enable Syntax highlighting
- boolean. Default: true

### markdown-pdf.highlightStyle

- Set the current `highlight.js` style file name. Examples: `github.css`, `monokai.css`, `base16/solarized-dark.css`
- [file name list](#)
- demo site : <https://highlightjs.org/demo>

```
"markdown-pdf.highlightStyle": "github.css",
```

## Markdown options

### markdown-pdf.breaks

- Enable line breaks
- boolean. Default: false

## Emoji options

### markdown-pdf.emoji

- Enable emoji. [EMOJI CHEAT SHEET](#)
- boolean. Default: true

## Configuration options

### markdown-pdf.executablePath

- Path to a Google Chrome, Microsoft Edge, or Chromium executable to run instead of the bundled Chromium
- See [How is the Chromium browser selected?](#) in the FAQ for how this setting interacts with installed browser detection and the managed Chromium download
- All `\` need to be written as `\\` (Windows)
- To apply the settings, you need to restart Visual Studio Code

```
"markdown-pdf.executablePath": "C:\\Program Files  
(x86)\\Google\\Chrome\\Application\\chrome.exe"
```

## Common Options

### markdown-pdf.scale

- Scale of the page rendering
- number. Default: 1

```
"markdown-pdf.scale": 1
```

## PDF options

- pdf only. [puppeteer page.pdf options](#)

### markdown-pdf.displayHeaderFooter

- Enables header and footer display
- boolean. Default: true
- Activating this option will display both the header and footer
- If you wish to display only one of them, remove the value for the other
- To hide the header

```
"markdown-pdf.headerTemplate": "",
```

- To hide the footer

```
"markdown-pdf.footerTemplate": "",
```

### markdown-pdf.headerTemplate

- Specifies the HTML template for outputting the header
- To use this option, you must set `markdown-pdf.displayHeaderFooter` to `true`
- `<span class='date'></span>` : formatted print date. The format depends on the environment
- `<span class='title'></span>` : markdown file name
- `<span class='url'></span>` : markdown full path name
- `<span class='pageNumber'></span>` : current page number
- `<span class='totalPages'></span>` : total pages in the document
- `%%ISO-DATETIME%%` : current date and time in ISO-based format (YYYY-MM-DD hh:mm:ss)
- `%%ISO-DATE%%` : current date in ISO-based format (YYYY-MM-DD)
- `%%ISO-TIME%%` : current time in ISO-based format (hh:mm:ss)
- Default (version 1.5.0 and later): Displays the Markdown file name and the date using `%%ISO-DATE%%`

```
"markdown-pdf.headerTemplate": "<div style=\"font-size: 9px; margin-left: 1cm;\">
<span class='title'></span></div> <div style=\"font-size: 9px; margin-left: auto;
margin-right: 1cm; \">%%ISO-DATE%%</div>",
```

- Default (version 1.4.4 and earlier): Displays the Markdown file name and the date using `<span class='date'></span>`

```
"markdown-pdf.headerTemplate": "<div style=\"font-size: 9px; margin-left: 1cm;\">
<span class='title'></span></div> <div style=\"font-size: 9px; margin-left: auto;
margin-right: 1cm; \"> <span class='date'></span></div>",
```

### markdown-pdf.footerTemplate

- Specifies the HTML template for outputting the footer
- For more details, refer to [markdown-pdf.headerTemplate](#)
- Default: Displays the {current page number} / {total pages in the document}

```
"markdown-pdf.footerTemplate": "<div style=\"font-size: 9px; margin: 0 auto;\">
<span class='pageNumber'></span> / <span class='totalPages'></span></div>",
```

### markdown-pdf.printBackground

- Print background graphics
- boolean. Default: true

### markdown-pdf.orientation

- Paper orientation
- portrait or landscape
- Default: portrait

### markdown-pdf.pageRanges

- Paper ranges to print, e.g., '1-5, 8, 11-13'
- Default: all pages

```
"markdown-pdf.pageRanges": "1,4-",
```

### markdown-pdf.format

- Paper format
- Letter, Legal, Tabloid, Ledger, A0, A1, A2, A3, A4, A5, A6
- Default: A4

```
"markdown-pdf.format": "A4",
```

### markdown-pdf.width

### markdown-pdf.height

- Paper width / height, accepts values labeled with units(mm, cm, in, px)
- If it is set, it overrides the markdown-pdf.format option

```
"markdown-pdf.width": "10cm",  
"markdown-pdf.height": "20cm",
```

### markdown-pdf.margin.top

### markdown-pdf.margin.bottom

### markdown-pdf.margin.right

### markdown-pdf.margin.left

- Paper margins.units(mm, cm, in, px)

```
"markdown-pdf.margin.top": "1.5cm",  
"markdown-pdf.margin.bottom": "1cm",
```

```
"markdown-pdf.margin.right": "1cm",  
"markdown-pdf.margin.left": "1cm",
```

## PNG, JPEG options

- png and jpeg only. [puppeteer page.screenshot options](#)

### markdown-pdf.quality

- jpeg only. The quality of the image, between 0-100. Not applicable to png images

```
"markdown-pdf.quality": 100,
```

### markdown-pdf.clip.x

### markdown-pdf.clip.y

### markdown-pdf.clip.width

### markdown-pdf.clip.height

- An object which specifies clipping region of the page
- number

```
// x-coordinate of top-left corner of clip area  
"markdown-pdf.clip.x": 0,  
  
// y-coordinate of top-left corner of clip area  
"markdown-pdf.clip.y": 0,  
  
// width of clipping area  
"markdown-pdf.clip.width": 1000,  
  
// height of clipping area  
"markdown-pdf.clip.height": 1000,
```

### markdown-pdf.omitBackground

- Hides default white background and allows capturing screenshots with transparency
- boolean. Default: false

## PlantUML options

### markdown-pdf.plantumlOpenMarker

- Opening delimiter used for the plantuml parser.



- Default: @startuml

### markdown-pdf.plantumlCloseMarker

- Closing delimiter used for the plantuml parser.
- Default: @enduml

### markdown-pdf.plantumlServer

- Plantuml server. e.g. http://localhost:8080
- Default: http://www.plantuml.com/plantuml
- For example, to run Plantuml Server locally [#139](#) :

```
docker run -d -p 8080:8080 plantuml/plantuml-server:jetty
```

[plantuml/plantuml-server - Docker Hub](#)

## markdown-it-include options

### markdown-pdf.markdown-it-include.enable

- Enable markdown-it-include.
- boolean. Default: true

## mermaid options

### markdown-pdf.mermaidServer

- mermaid server
- Default: https://unpkg.com/mermaid/dist/mermaid.min.js

## FAQ

### How can I change emoji size ?

1. Add the following to your stylesheet which was specified in the markdown-pdf.styles

```
.emoji {  
  height: 2em;  
}
```

## Auto guess encoding of files

Using [files.autoGuessEncoding](#) option of the Visual Studio Code is useful because it automatically guesses the character code. See [files.autoGuessEncoding](#)

```
"files.autoGuessEncoding": true,
```

## Output directory

If you always want to output to the relative path directory from the Markdown file.

For example, to output to the "output" directory in the same directory as the Markdown file, set it as follows.

```
"markdown-pdf.outputDirectory" : "output",  
"markdown-pdf.outputDirectoryRelativePathFile": true,
```

## Page Break

Please use the following to insert a page break.

```
<div class="page"/>
```

## Why did my heading anchors change?

Starting with 2.0.0, Markdown PDF generates heading IDs using a custom `markdown-it-named-headers` implementation that follows GitHub-compatible VS Code slug generation. Compared to the previous implementation, the new slug generator preserves CJK characters and underscores while removing unsupported punctuation, which can cause existing internal anchors (e.g. `#some-heading`) to resolve differently.

If your Markdown relies on specific anchor strings (for example, a table of contents or cross-document links), re-check the generated anchors after exporting and update the links as needed.

## Why did my syntax highlight style stop working?

Starting with 2.0.0, Markdown PDF uses `highlight.js` v11 (previously v9). Some style names from v9 have been renamed or removed. Markdown PDF maps legacy style names to current names where possible and shows a warning message when a configured style cannot be found. If no mapping is available, the extension falls back to `tomorrow.css`.

Please check the [available styles](#) and update your `markdown-pdf.highlightStyle` setting to a current style name.

## Why is my front matter no longer parsed?

Starting with 2.0.0, Markdown PDF parses YAML front matter with a custom implementation instead of `gray-matter`. The new parser is stricter and rejects the following cases that the old parser may have accepted:

- Top-level YAML sequences (arrays) as front matter
- Front matter that does not parse into a plain object

- Malformed YAML structures

A valid front matter must be a YAML mapping (object) at the top level, for example:

```
---
title: My Document
"markdown-pdf":
  displayHeaderFooter: true
---
```

BOM-prefixed files are still supported.

How is the Chromium browser selected?

Markdown PDF resolves a Chromium-based browser in the following order:

1. The path specified in `markdown-pdf.executablePath`, if the file exists.
2. An installed browser on your system. Google Chrome (stable) is detected via `@puppeteer/browsers` at its standard OS install location; Microsoft Edge and Chromium are probed at the fixed paths listed below.
3. A managed Chromium that Markdown PDF automatically downloads on first use.

The first match wins. The per-OS scan order for installed Edge and Chromium is:

## Windows

1. Google Chrome (stable install, detected via `@puppeteer/browsers`)
2. `%LOCALAPPDATA%\Microsoft\Edge\Application\msedge.exe`
3. `%LOCALAPPDATA%\Chromium\Application\chrome.exe`
4. `%PROGRAMFILES%\Microsoft\Edge\Application\msedge.exe`
5. `%PROGRAMFILES%\Chromium\Application\chrome.exe`
6. `%PROGRAMFILES(X86)%\Microsoft\Edge\Application\msedge.exe`
7. `%PROGRAMFILES(X86)%\Chromium\Application\chrome.exe`

## macOS

1. Google Chrome (stable install, detected via `@puppeteer/browsers`)
2. `/Applications/Chromium.app/Contents/MacOS/Chromium`
3. `/Applications/Microsoft Edge.app/Contents/MacOS/Microsoft Edge`

## Linux

1. Google Chrome (stable install, detected via `@puppeteer/browsers`)
2. `/usr/bin/chromium-browser`
3. `/usr/bin/chromium`
4. `/usr/bin/microsoft-edge`
5. `/usr/bin/microsoft-edge-stable`

Where is Chromium downloaded?

If no installed browser is found, Markdown PDF downloads a managed Chromium on first use. The download is stored under the extension's VS Code global storage directory:

| OS      | Download path                                                             |
|---------|---------------------------------------------------------------------------|
| Windows | %APPDATA%\Code\User\globalStorage\yzane.markdown-pdf\                     |
| macOS   | ~/Library/Application Support/Code/User/globalStorage/yzane.markdown-pdf/ |
| Linux   | ~/.config/Code/User/globalStorage/yzane.markdown-pdf/                     |

If you use VS Code Insiders or VSCodium, the base path changes accordingly (for example `Code - Insiders` or `VSCodium` instead of `Code`).

During the download, `Installing Chromium` is shown in the status bar.

## Known Issues

### `markdown-pdf.styles` option

- Online CSS (<https://xxx/xxx.css>) is applied correctly for JPG and PNG, but problems occur with PDF. [#67](#)

## Release Notes

### 2.0.0 (2026/04/11)

- Breaking: Heading ID slug generation, front matter parsing, and Chromium resolution have changed. See the [FAQ](#) for details.
- Change: Migrate to TypeScript and bundle with esbuild
- Change: Bundle `puppeteer-core` and manage Chromium via the built-in `chromium-resolver` (installed Chrome/Edge preferred, auto-download fallback)
- Change: Replace `markdown-it-include`, `markdown-it-named-headers`, and `markdown-it-checkbox` with in-repo custom implementations
- Change: Remove `cheerio`, `mustache`, and `gray-matter` dependencies
- Add: Unit and integration test suites (`vscode-test-cli`)

For details, see [Change Log](#).

### 1.6.0 (2025/04/15)

- Refactor: replace external checkbox and named-header markdown-it packages with built-in implementations
- Fix: Allow underscores in section header identifiers [#404](#)
- Update: use GitHub-compatible [VS Code slug generation](#) for heading identifiers

## License

MIT

## Special thanks

- [puppeteer/puppeteer](#)
- [markdown-it/markdown-it](#)
- [markdown-it/markdown-it-emoji](#)
- [HenrikJoreteg/emoji-images](#)
- [highlightjs/highlight.js](#)
- [markdown-it/markdown-it-container](#)
- [gmunguia/markdown-it-plantuml](#)
- [mermaid-js/mermaid](#)